ARJUNA JEE 2027

Lecture -02 | Mathematics

PERMUTATIONS AND COMBINATIONS

By- Sachin Jakhar Sir



Topics *to be covered*

A **Formation of Numbers**

B **Question Practice**

C

D



H.W. Discussion

Question [JEE (Adv.)-2011]

The positive integer value of $n > 3$ satisfying the equation

$$\frac{1}{\sin\left(\frac{\pi}{n}\right)} = \frac{1}{\sin\left(\frac{2\pi}{n}\right)} + \frac{1}{\sin\left(\frac{3\pi}{n}\right)}$$ is

$\# \quad \frac{\pi}{n} = \theta$

$$\frac{1}{\sin\theta} = \frac{1}{\sin 2\theta} + \frac{1}{\sin 3\theta}$$

$$\frac{1}{\sin\theta} - \frac{1}{\sin 3\theta} = \frac{1}{\sin 2\theta}$$

$$\frac{\overset{C}{\sin 3\theta} - \overset{D}{\sin\theta}}{\sin\theta \sin 3\theta} = \frac{1}{2\sin\theta\cos\theta}$$

$$2\cos 2\theta \sin\theta \cdot 2\cos\theta = \sin 3\theta$$

$$\sin 2\theta$$

$$\sin 4\theta = \sin 3\theta$$

$\sin 4\theta = \sin 3\theta$

$4\theta = n\pi + (-1)^n 3\theta$

$n \to$ even

$4\theta = n\pi + 3\theta$

$\theta = n\pi$

$\frac{\pi}{n} = n\pi \rightarrow n = \frac{1}{m} = \frac{1}{2}, \frac{1}{4}, \frac{1}{6} \ldots$

$n \to$ odd

$4\theta = n\pi - 3\theta$

$7\theta = n\pi$

$7\frac{\pi}{n} = n\pi$

$\frac{7}{n} = n$

$m = 1 \rightarrow n = 7$

$m = 3 \rightarrow n = \frac{7}{3}$ X

$m = 5 \rightarrow \frac{7}{5} = n$ X

$m = 7 \rightarrow n = 1$

Example

Draw following graphs using multiple transformation:

(i) $y = |\ln|x||$

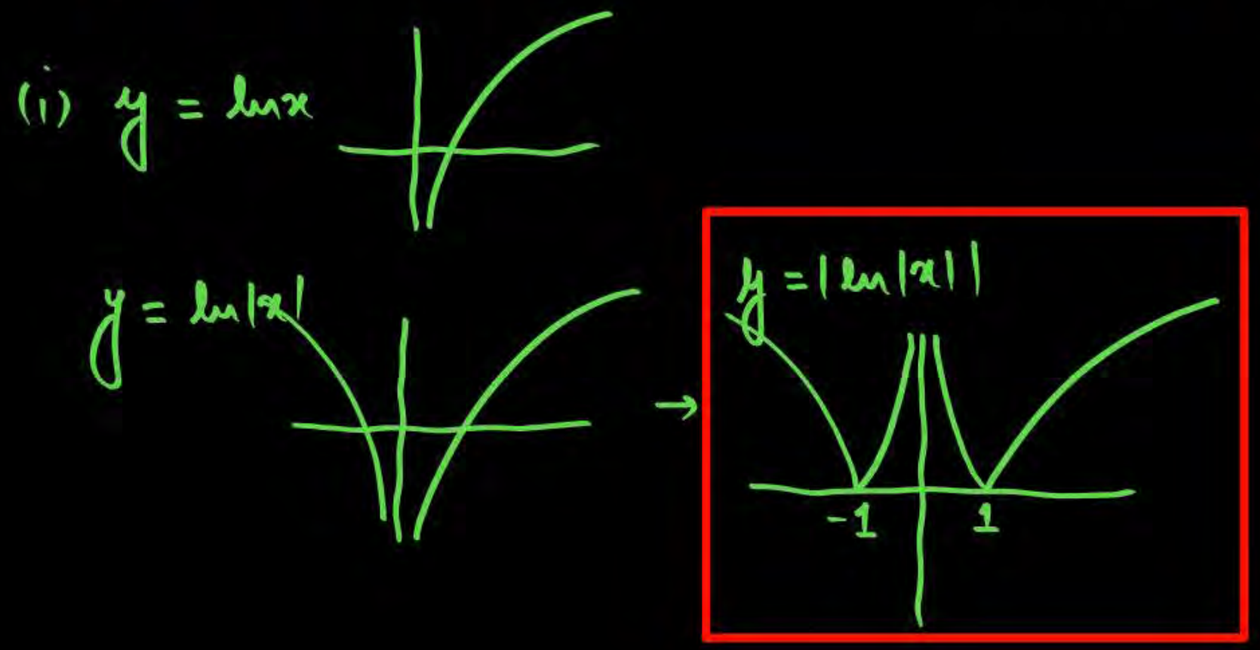
(ii) $y = \big||x| - 2\big| - 3\big|$

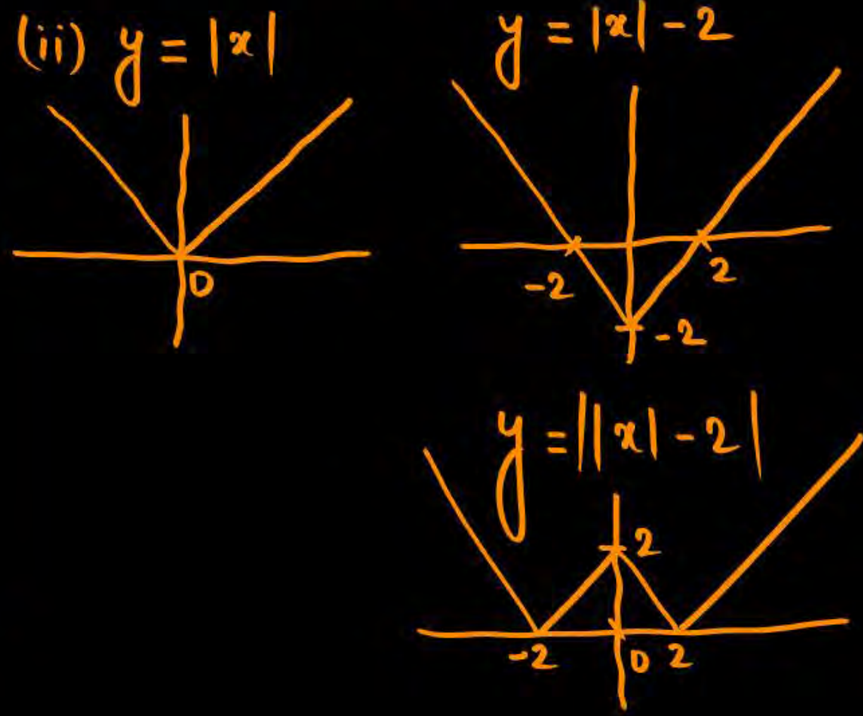
(iii) $y = e^{-|x|} + 2$

(iv) $y = -\{-x\}$

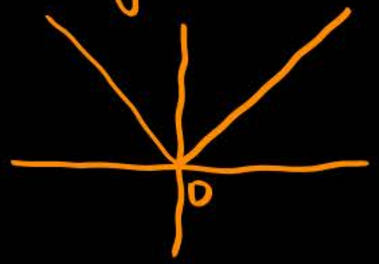
(v) $|y| = 3 + \sqrt{2 - x}$

(vi) $y = |x^2 - 5|x| + 6|$

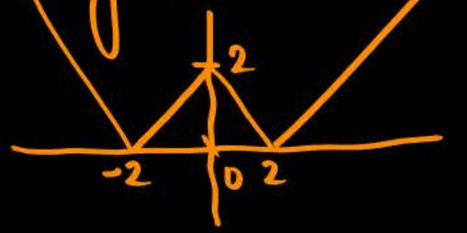




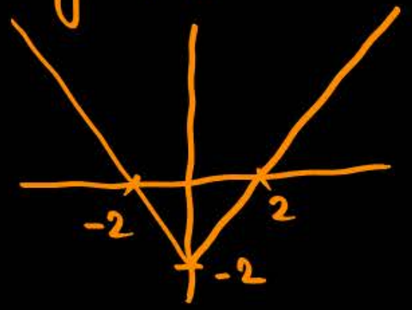
(ii) $y = |x|$



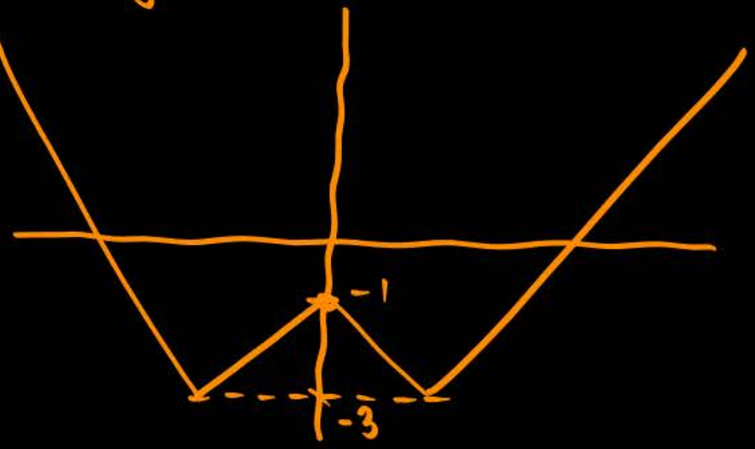
$y = ||x| - 2|$



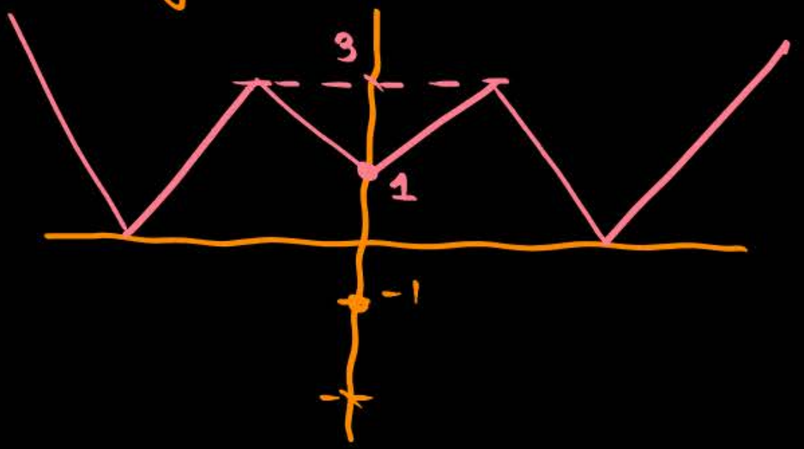
$y = |x| - 2$



$y = ||x| - 2| - 3$



$y = |||x| - 2| - 3|$



Example

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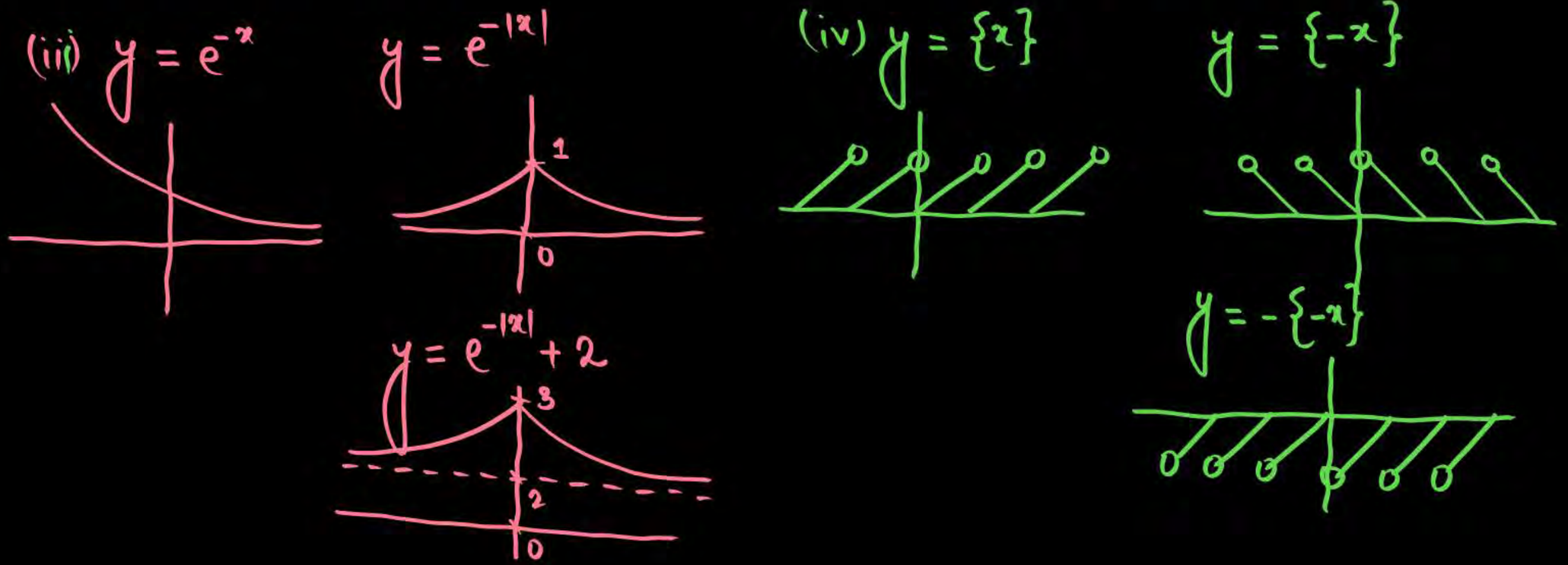
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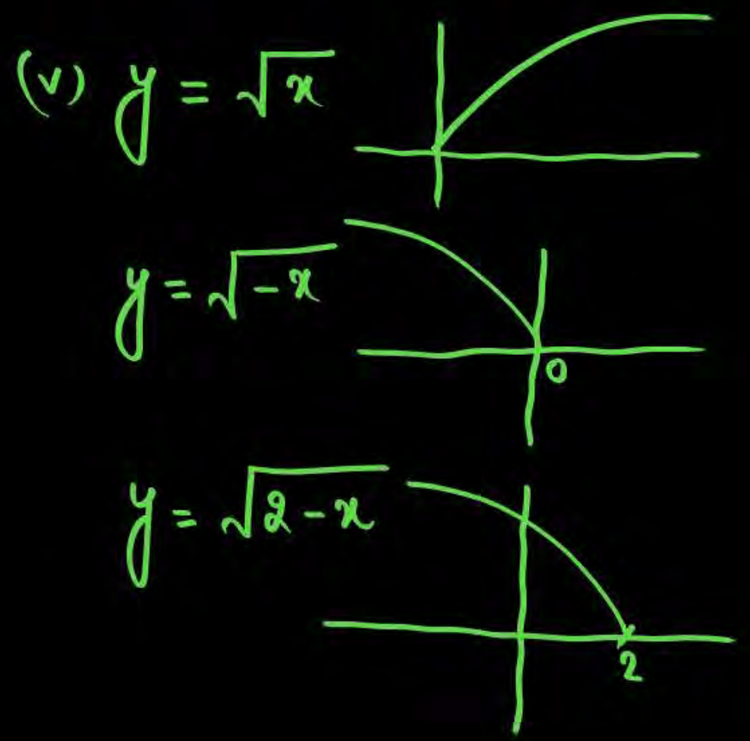
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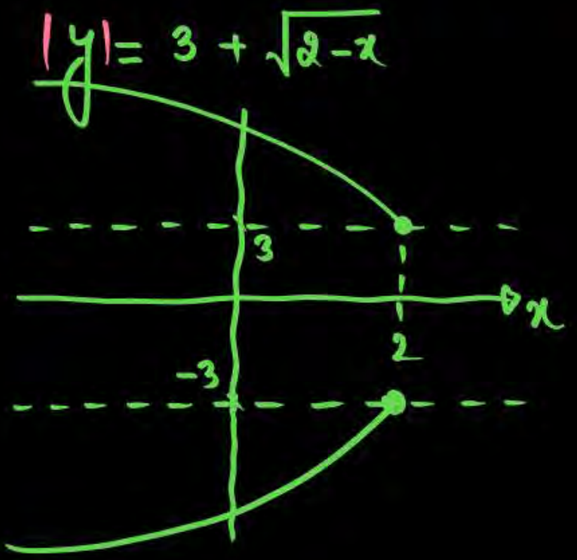
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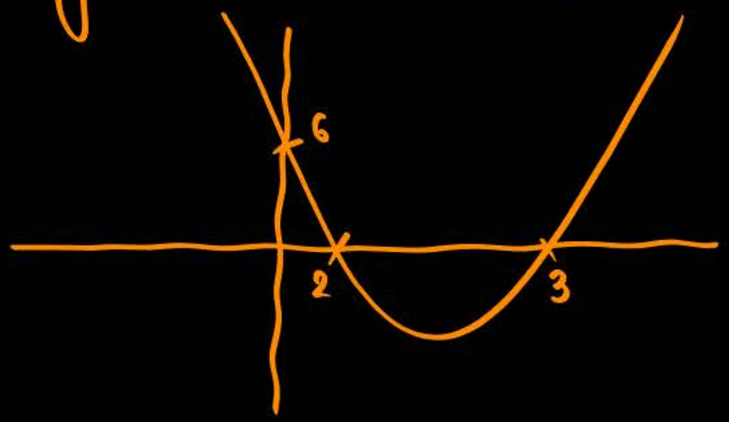
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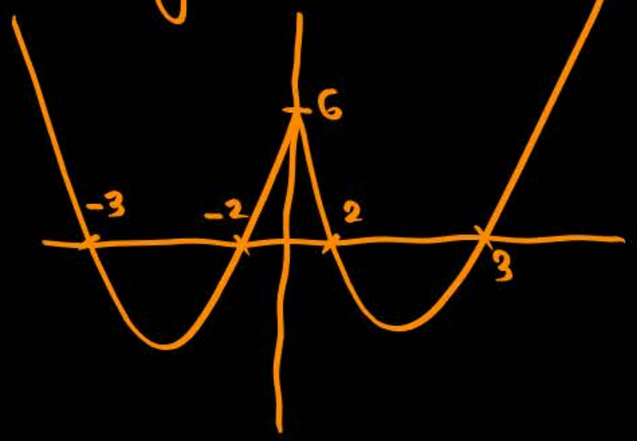
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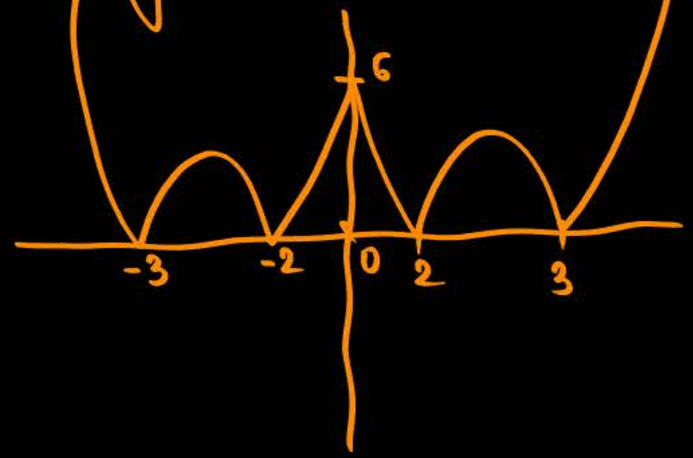
(vi) $y = x^2 - 5x + 6 = (x-2)(x-3)$



$y = x^2 - 5|x| + 6$

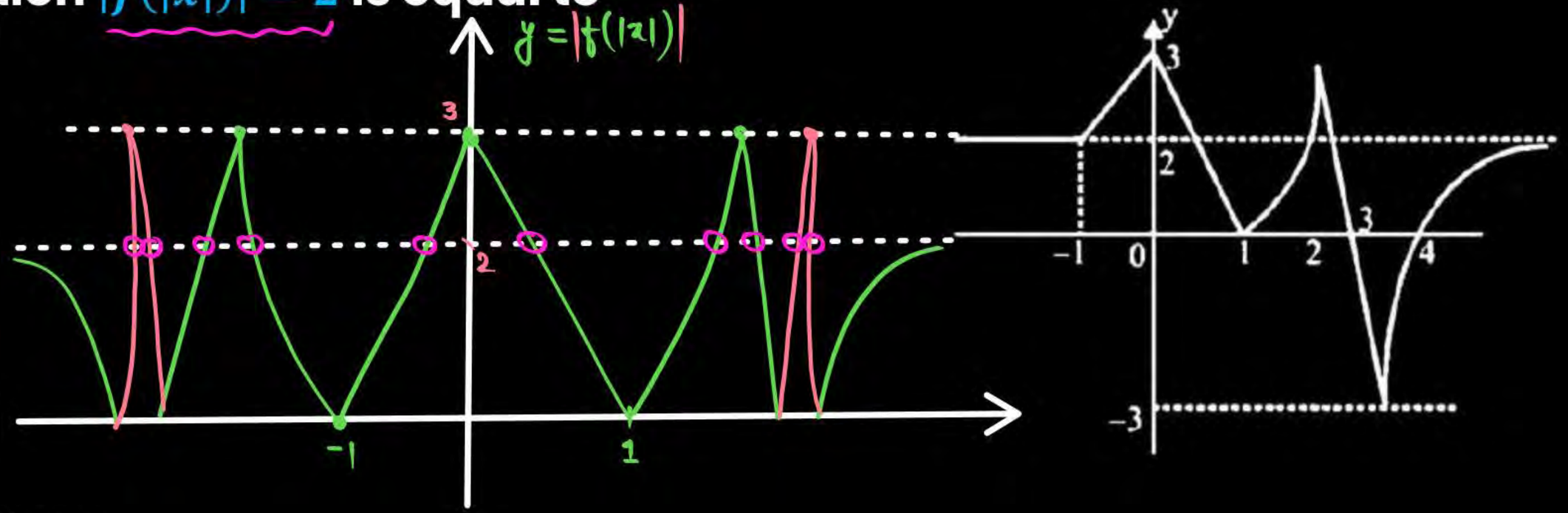


$y = |x^2 - 5|x| + 6|$



Question

If graph of a function $f(x)$ is shown as following then number of solutions of equation $|f(|x|)| = 2$ is equal to



A 3

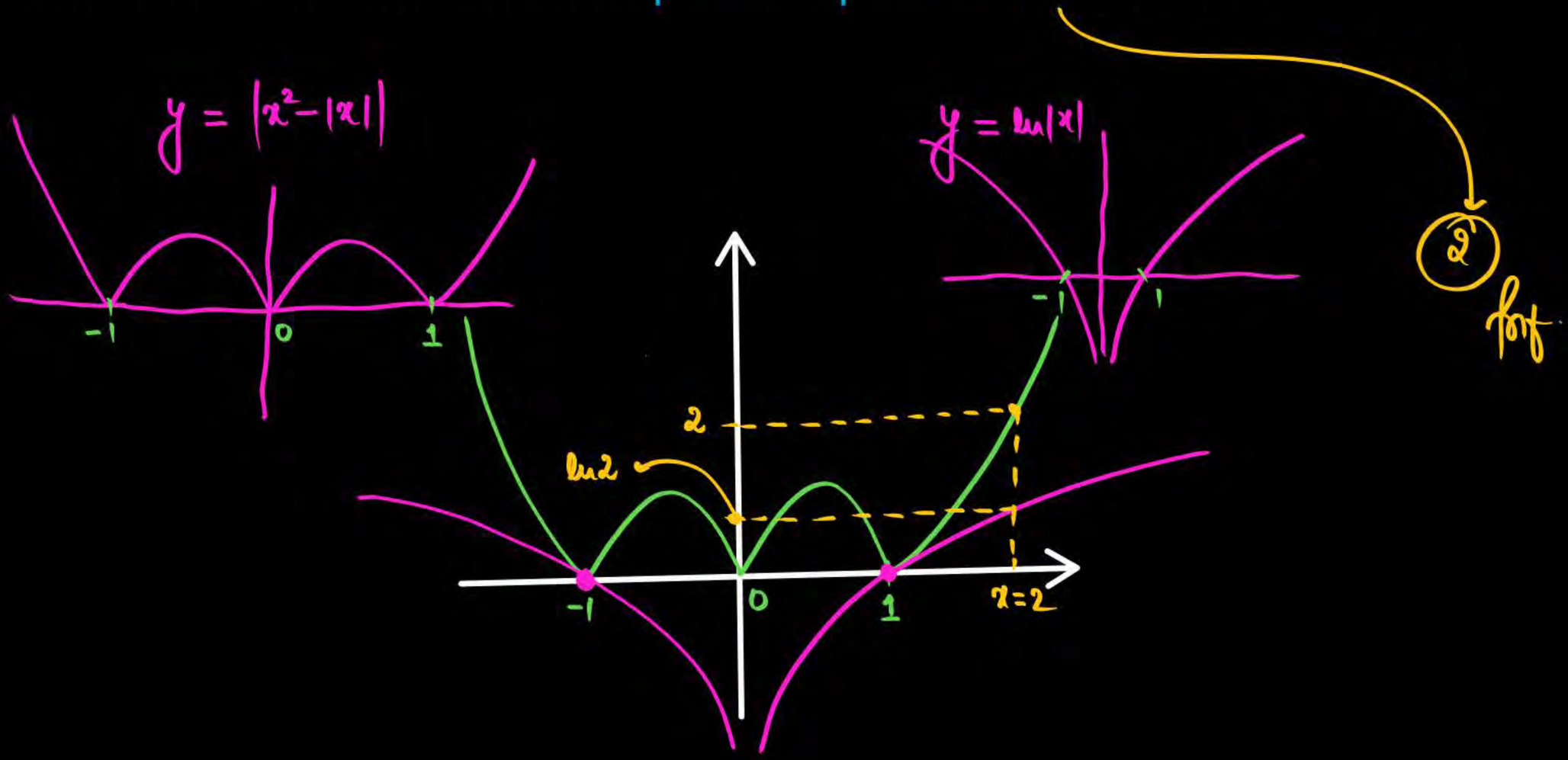
B 6

C 10

D Infinite

Question

Number of solutions of equation $\left|x^2 - |x|\right| = \ln|x|$ is equal to





MAX. & MIN. CONCEPT OF GRAPHS

(i) $\max.\{f(x), g(x)\} = \frac{f(x)+g(x)}{2} + \left|\frac{f(x)-g(x)}{2}\right|$

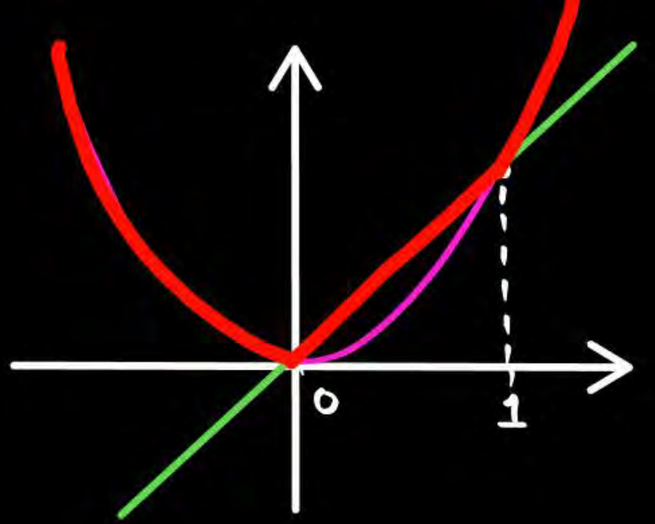
(ii) $\min.\{f(x), g(x)\} = \frac{f(x)+g(x)}{2} - \left|\frac{f(x)-g(x)}{2}\right|$

$\max.\{2, 3, 4, 5\} = 5.$

$y = \max.\{f(x), g(x)\}$

Example

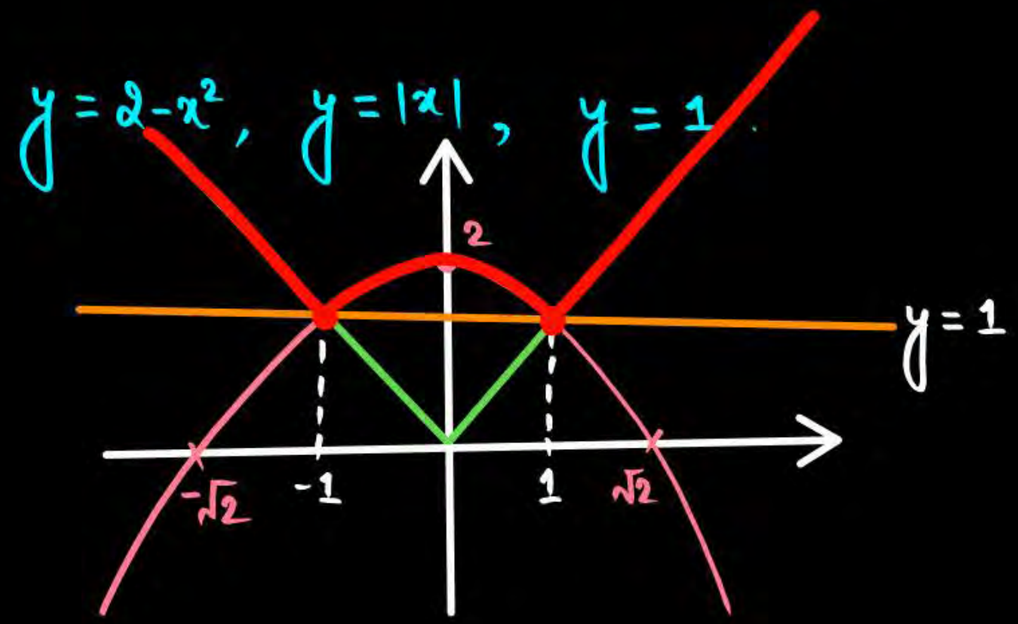
(i) Draw : $y = \max.\{x^2, x\}$

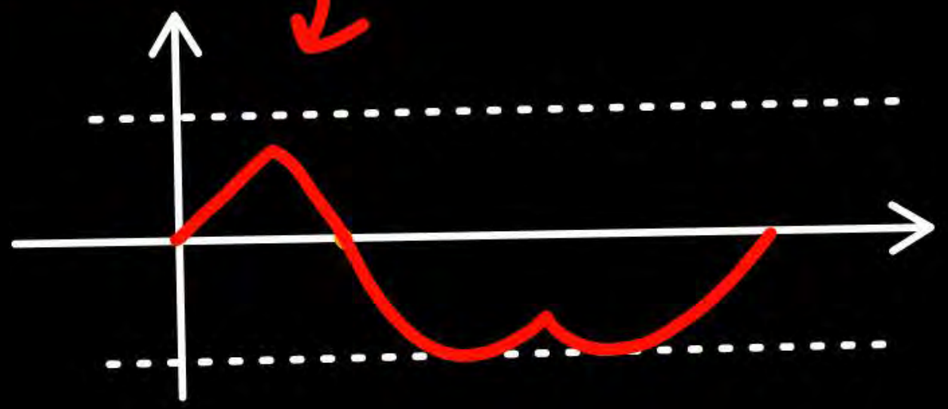


(ii) Draw : $y = \min.\{x^2, x\}$

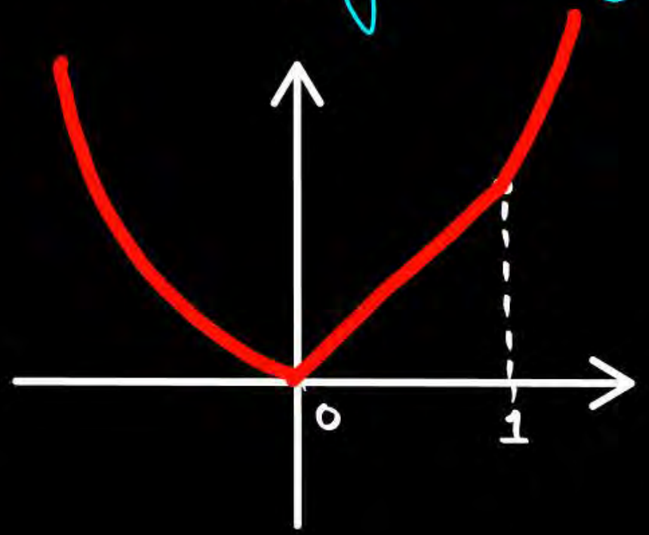
(iii) Draw : $y = \max.\{2 - x^2, |x|, 1\}$

(iv) Draw : $y = \min.\{\sin x, \cos x\}$ where $x \in [0, 2\pi]$

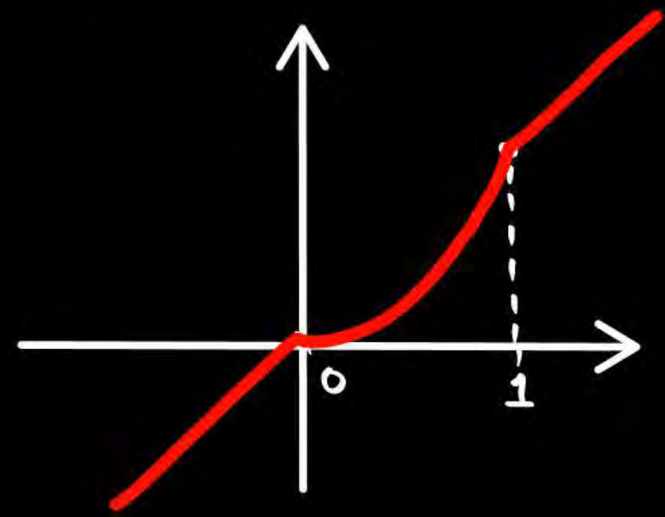




(i) $y = \max.\{x^2, x\}$



(ii) $y = \min.\{x^2, x\}$



Bumper Question

Find number of solution of following equations:



(i) $|\sin \pi x| = \ln x$

(ii) $\max.\{\sin x, \cos x\} = \frac{\sqrt{3}}{2}, x \in [0, 100\pi]$

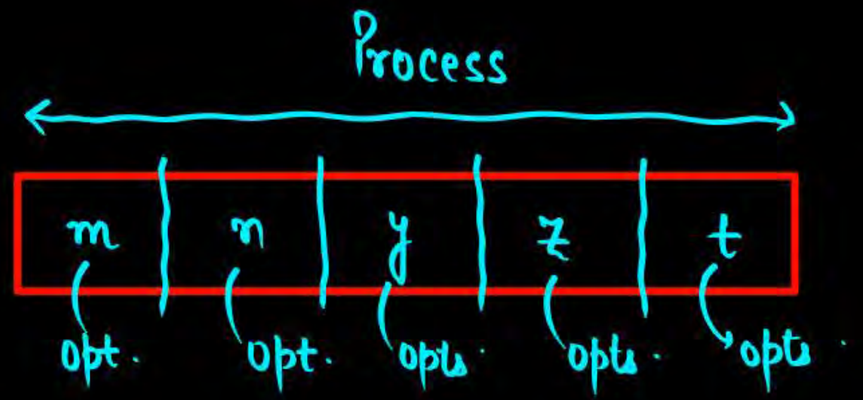
(iii) $\min.\{\tan x, \cot x\} = \frac{1}{8}, x \in (-2\pi, 2\pi)$

(iv) $2\sin|x| - |x| + 1 = 0$

(v) $3|\cos x| = 5|\sin x|, x \in [0, 3\pi]$

PERMUTATIONS & COMBINATIONS

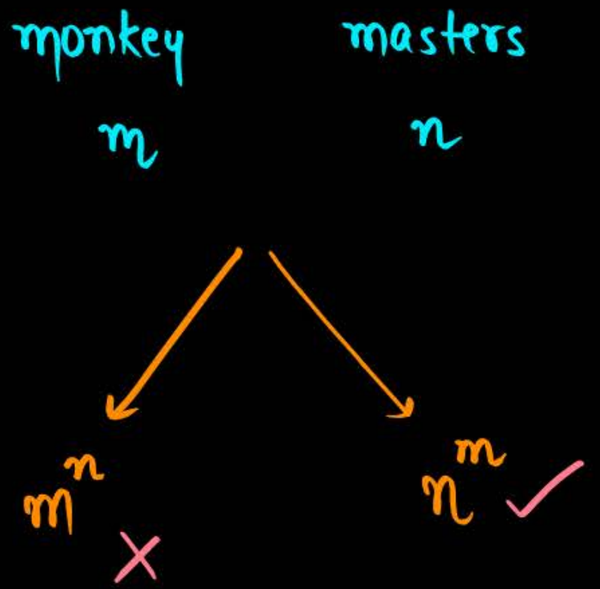
LAST CLASS RECAP

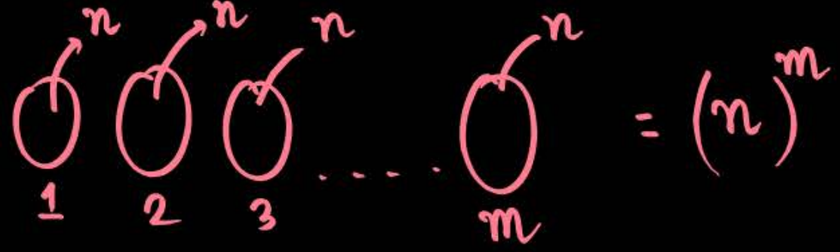


Total ways = m n y z t .


P1
P2
P3


3 3 . . . . .
= 3^6


monkey
m
masters
n
m^n
×
n^m


1 2 3 . . . . m
= (n)^m

Question

[Ans. : B]

10 students compete in a swimming race. In how many ways can they occupy the first 3 positions?

A 120

B 720

C 504

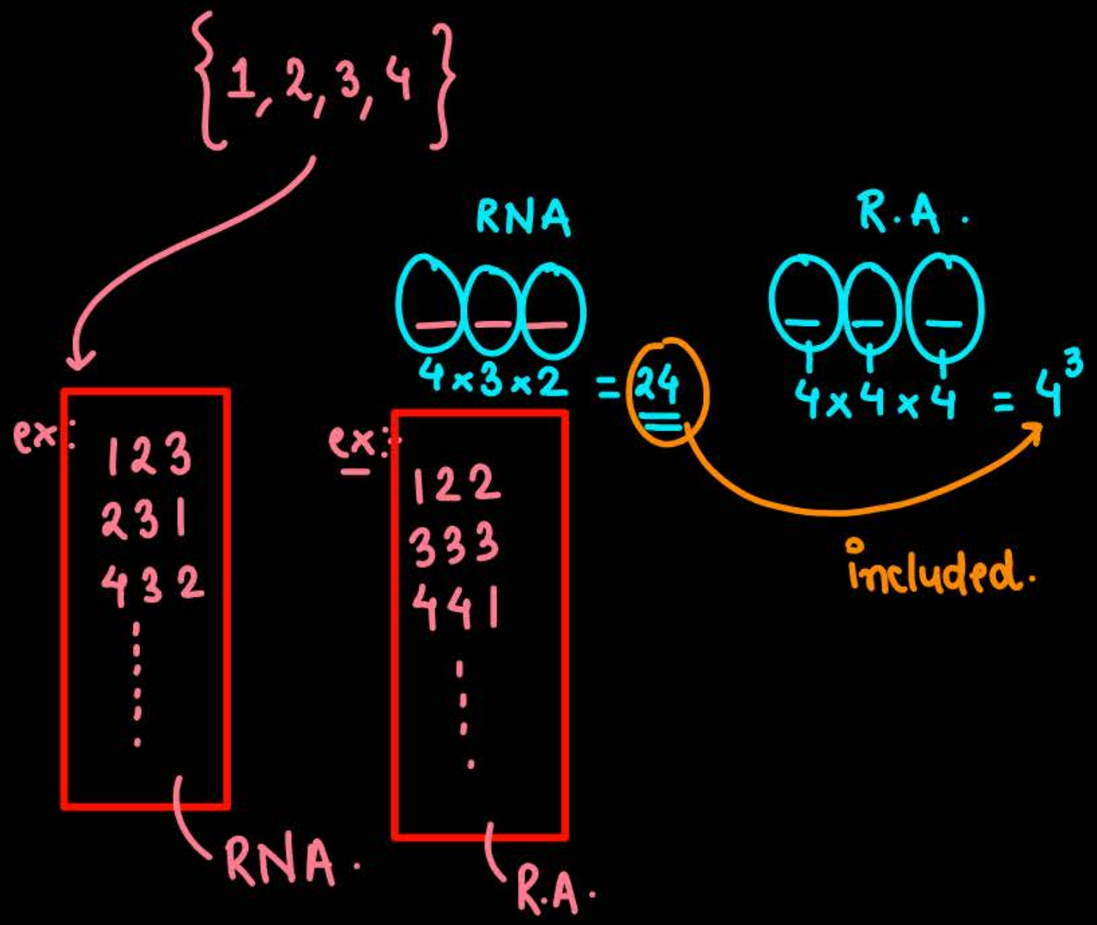
D 1000

1st 2nd 3rd

$10 \times 9 \times 8 = 720$

FORMATION OF NUMBERS

100% Ques?.



{0,1,2,3}

RNA _ _ _ = 18
3 3 2

RA _ _ _ = 48
3 4 4

Question

Number of different five digit numbers that can be formed using digits {0, 1, 2, 7, 8, 9} such that:

(i) Repetition of digits is not allowed(R.N.A.).

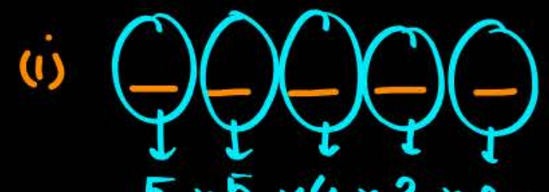
(ii) Repetition of digit is allowed (R.A.).

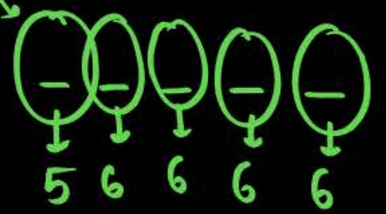
(iii) Number formed is even (R.N.A)

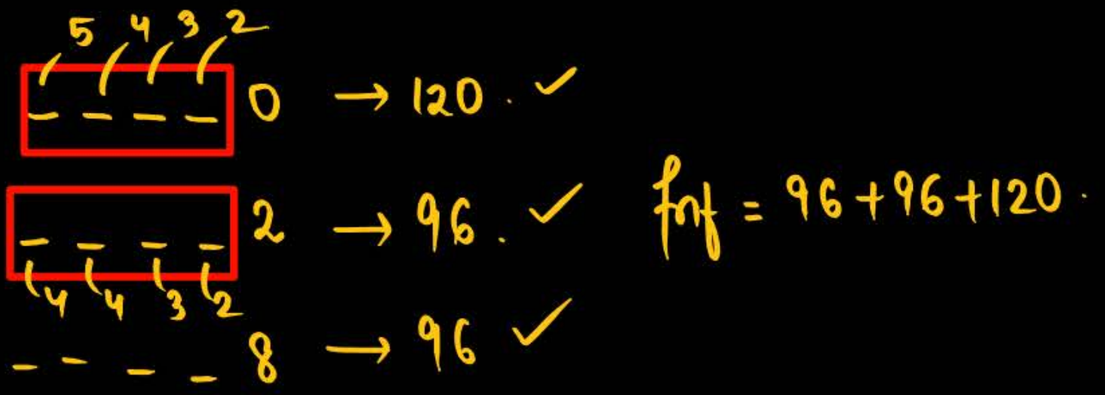
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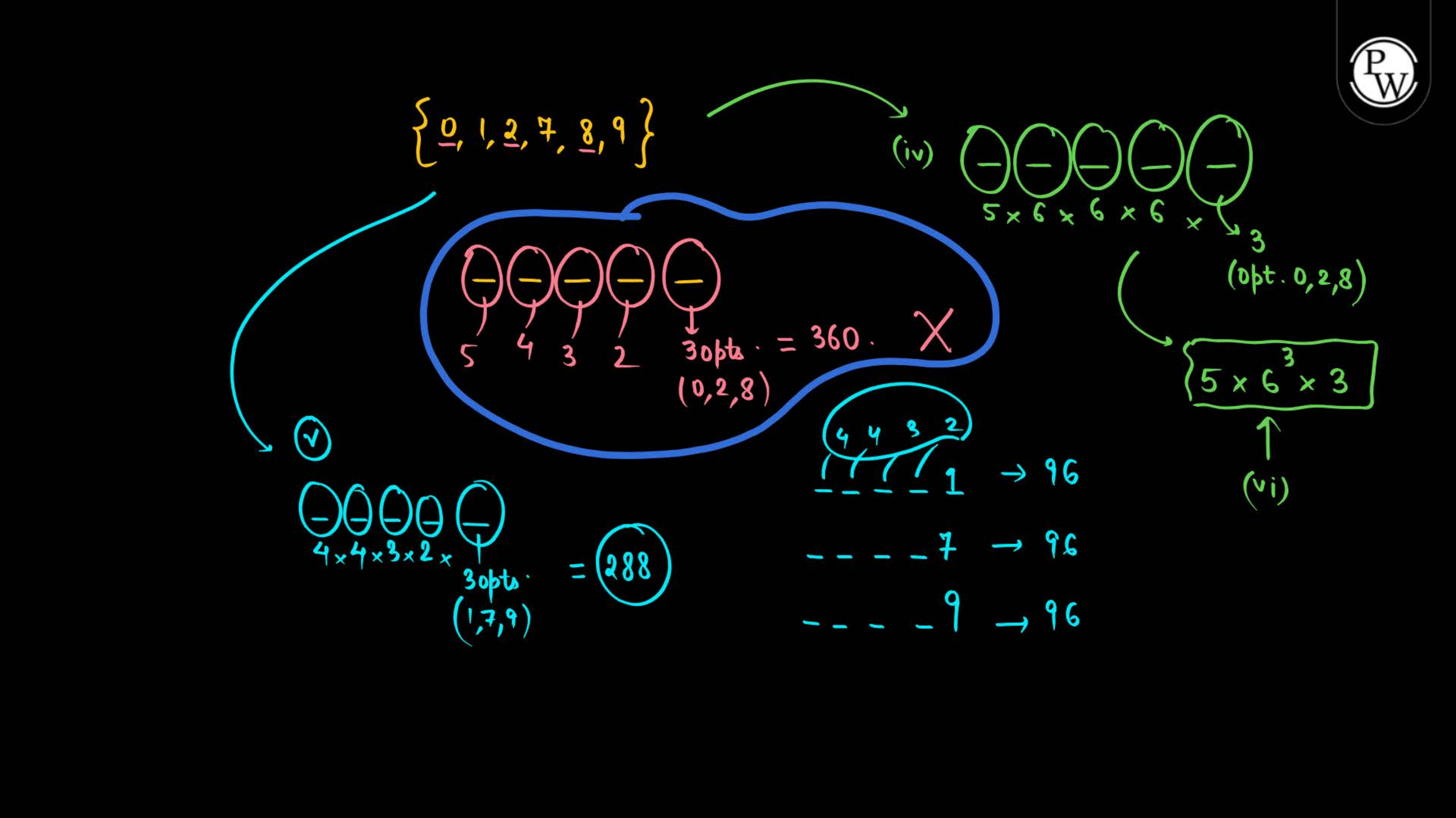
(v) Number formed is odd (R.N.A.)

(vi) Number formed is odd (R.A.)








{0, 1, 2, 7, 8, 9}
(iv)
5 × 6 × 6 × 6 ×
3
(0pt. 0,2,8)
5 4 3 2
30pts. = 360. X
(0,2,8)
5 × 6 × 6 × 6 × 3
(vi)
(v)
4 × 4 × 3 × 2 ×
30pts.
(1,7,9)
= 288
4 4 3 2
1 → 96
7 → 96
9 → 96

Question [JEE Mains-2015]

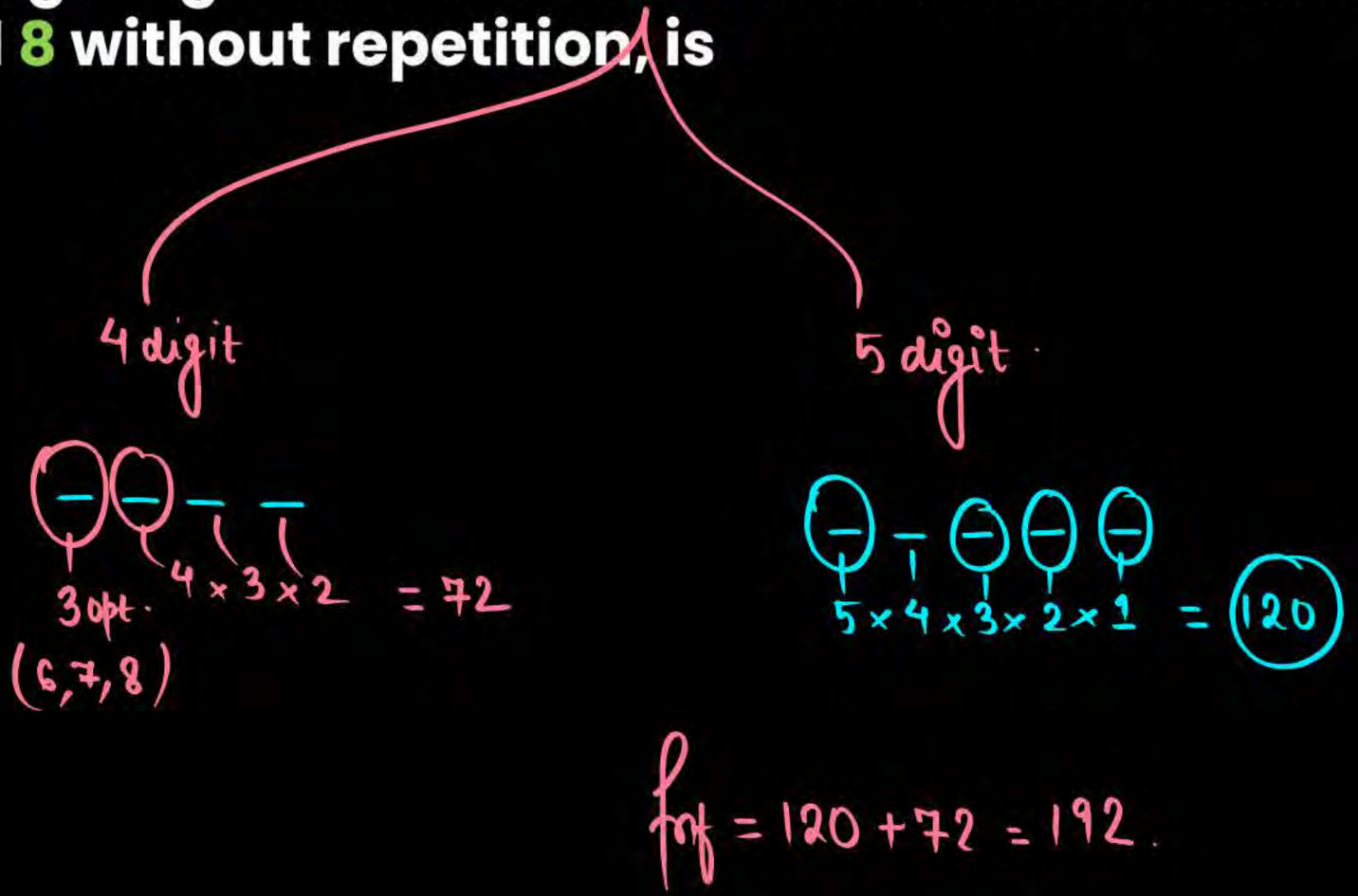
The number of integers greater than 6000 that can be formed, using the digits 3, 5, 6, 7 and 8 without repetition, is

A 120

B 72

C 216

D 192



Question [JEE Mains-2019]

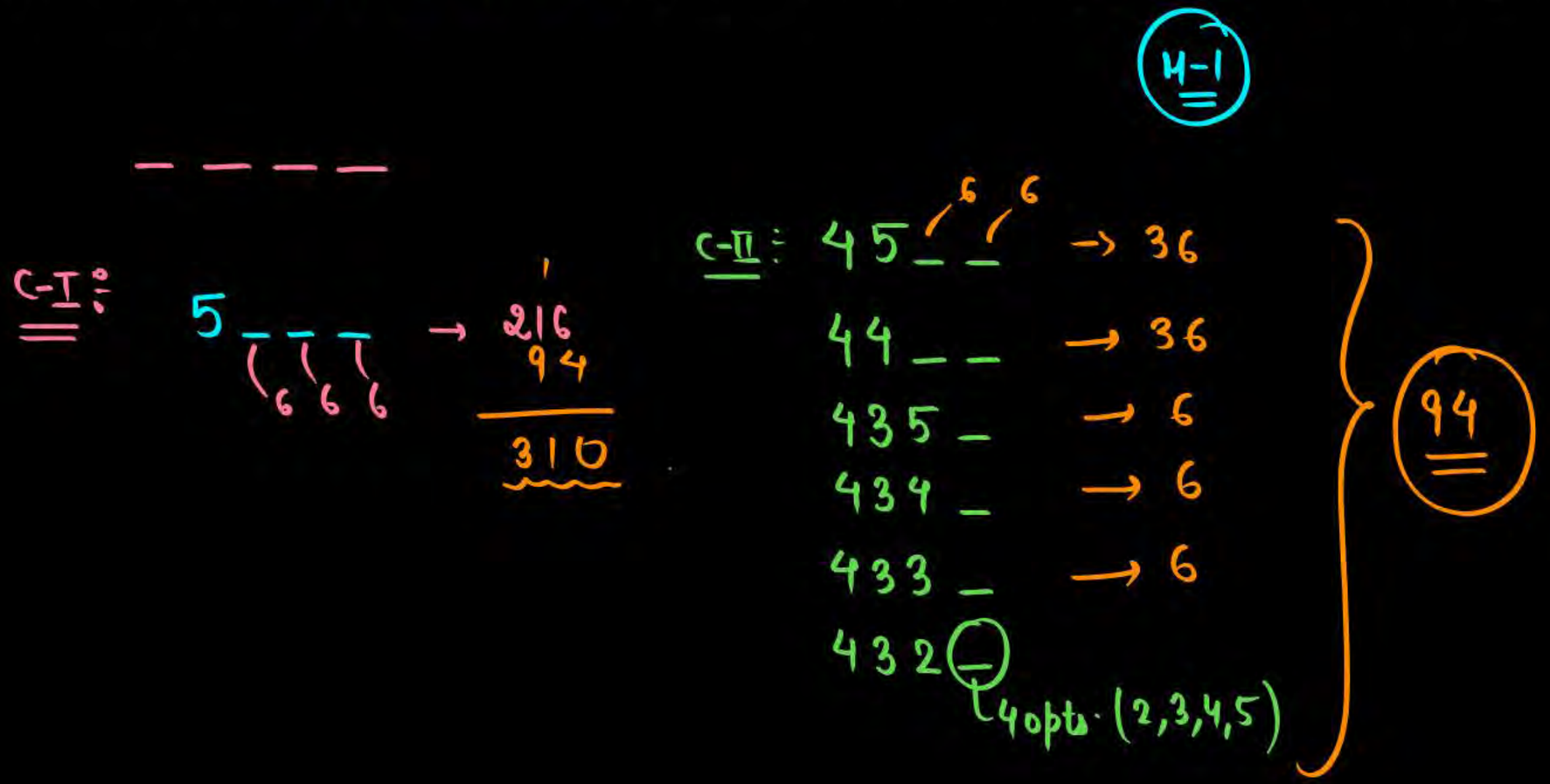
The number of four-digit numbers strictly greater than 4321 that can be formed using the digits 0, 1, 2, 3, 4, 5 (repetition of digits is allowed) is

A 360

B 306

C 310

D 288



Question [JEE Mains-2019]

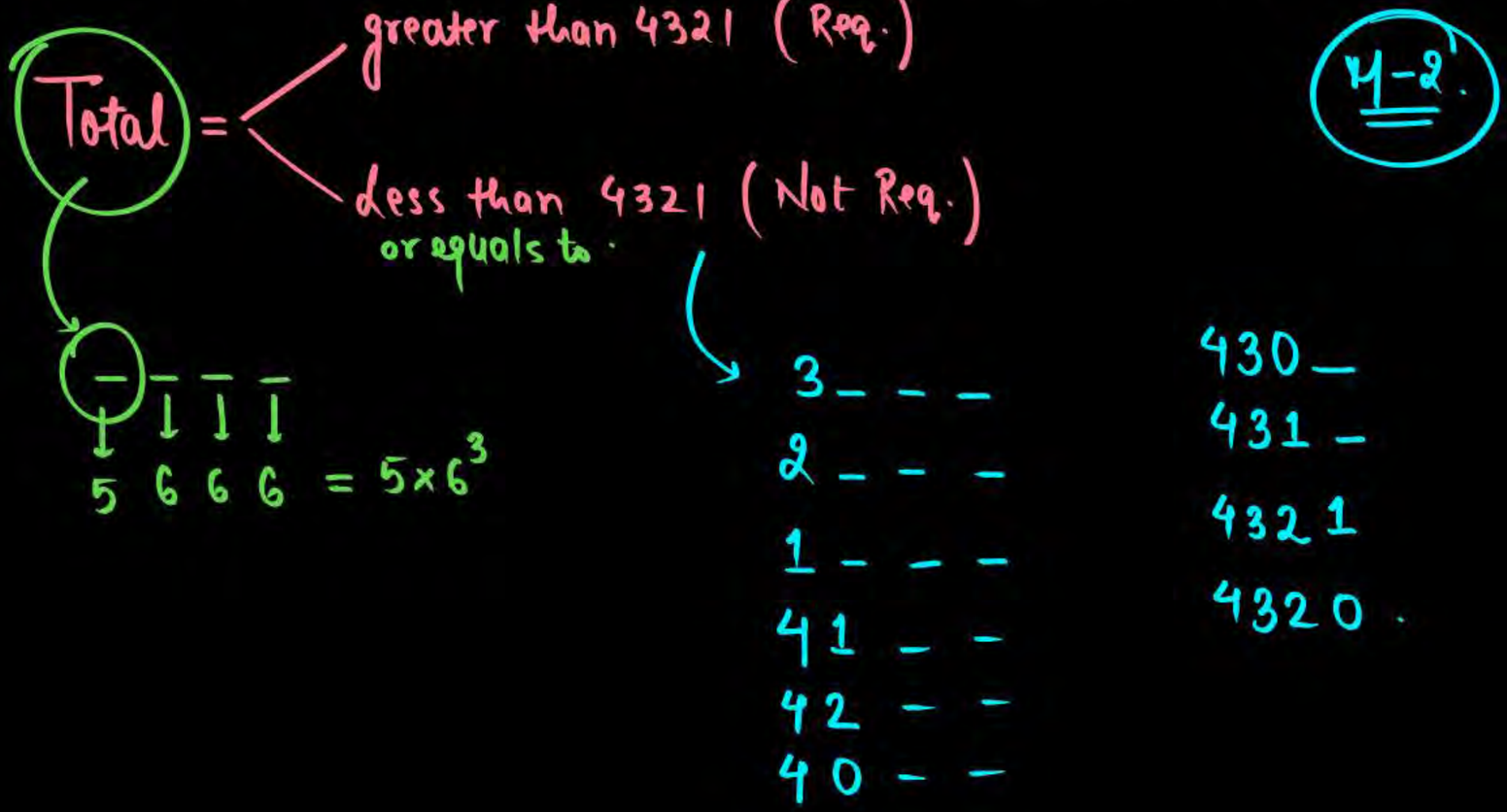
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Blue बादल (KKPNCC)

जब Required cases ज्यादा हों तो ek बार

Not required cases के बारे में सोचो।

BIJECTION: Required = Total - (Not Required)

DO IT BY YOURSELF (DIBY)

DIBY-01

A man has 3 jackets, 10 shirts and 5 pairs of slacks. If an outfit consists of a jacket, a shirt, and a pair of slacks, how many different outfits can the man make?

[Ans. : 150]

DIBY-02

There are 6 roads between A & B and 4 roads between B & C.

(i) In how many ways can one drive from A to C by way of B?

(ii) In how many ways can one drive from A to C and back to A, passing through B on both trips?

(iii) In how many ways can one drive the circular trip described in (ii) without using the same road more than once.

[Ans. : (i) 24; (ii) 576; (iii) 360]

DIBY-03

If repetitions are not permitted

(i) How many 3 digit numbers can be formed from the six digits 2, 3, 5, 6, 7 & 9?

(ii) How many of these are less than 400?

(iii) How many are even?

(iv) How many are odd?

(v) How many are multiple of 5?

[Ans. : (i) 120; (ii) 40; (iii) 40; (iv) 80; (v) 20]

DIBY-04

How many natural numbers are their from 1 to 1000 which have none of their digits repeated.

[Ans. : 738]



Homework

Re-attempt all Class Examples

DPP & DIBY

It is never about end result

It's always about Journey

Thank You

#futureIITians